CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Varvak, Edward Ariel

PERSISTENT IDENTIFIER (PID) OF THE SENIOR/KEY PERSON: https://orcid.org/0009-0004-8413-5873

*POSITION TITLE: Graduate Student

*ORGANIZATION AND LOCATION: University of Chicago, Chicago, Illinois, United States

Proposals/Active Projects

***Proposal/Active Project Title:**	MSPRF: The Hodge theory of Lagrangian fibrations, with applications to hyper-Kahler geometry (this proposal)
***Status of Support:**	Pending
Proposal/Award Number:	
***Source of Support:**	National Science Foundation
***Primary Place of Performance:**	Massachusetts Institute of Technology, Cambridge, MA, USA
***Proposal/Active Project Start Date: (MM/YYYY):**	09/2027
***Proposal/Active Project End Date: (MM/YYYY):**	08/2031
***Total Anticipated Proposal/Project Amount:**	$190,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	8
2028	8
2029	8

***Overall Objectives:**

The project studies the Hodge theory of Lagrangian fibrations of hyper-Kahler manifolds and related holomorphic symplectic varieties. Its objectives are
(1) to determine which decompositions of the Mumford--Tate group occur for fibers of Lagrangian fibrations, and whether a decomposable fiber forces a hyper-Kahler manifold to be deformation equivalent to a Hilbert scheme of points on a K3 surface;
(2) to extend this Hodge theory to non-compact Lagrangian fibrations, chiefly the Hitchin fibration, by determining the Mumford--Tate group of its very general fiber; and
(3) to work toward the hyper-Kahler Bogomolov--Mumford conjecture by producing rational curves from degenerations of fibers with an elliptic factor.

***Statement of Potential Overlap:**

There is no potential overlap.

Certification:

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

In accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19232), each individual identified as a senior/key person must certify that they are not a party to a malign foreign talent recruitment program.

Research Security Training Requirement for Federal Award Personnel: In accordance with Section 10634 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), each individual identified as a senior/key person must certify that they have completed the requisite research security training that meets the requirements specified in Item 2 of Important Notice No. 149 within 12 months prior to proposal submission.

Certified by Varvak, Edward in SciENcv on 2026-10-20 19:26:32